

Building a Bidirectional Nepali Sign Language Translation System: Research, Open-Source, and Productization Roadmap

TL;DR

- **Build it as a Deaf-led research consortium, not a solo ML project.** No production-quality NSL corpus exists: the entire public NSL computational record is one continuous CSLR paper (40-sentence Shree Deaf School dataset, 20.23% WER, MediaPipe Holistic + Spatial-Temporal Fusion Transformer), the NSL23 alphabet dataset (Sunuwar et al., *Data in Brief* 2024), and a 54,000-image character benchmark (Poudel et al., arXiv:2510.11243, 2025). NSL is excluded from every major multilingual SLP/SLR effort (SignLLM/Prompt2Sign, YouTube-SL-25, Uni-Sign, SignAvatars). That gap is the opportunity.
- **The right architecture is pose-centric, gloss-optional, LLM-decoded recognition** (TwoStream-SLR / CorrNet+ encoders → SHuBERT/Uni-Sign self-supervised pretraining → Sign2GPT-style LLM decoder emitting Nepali Devanagari) and a 3D-pose-first production stack (SMPL-X target via SignAvatars-style fitting → diffusion or flow-matching motion generator → Blender/SMPL-X avatar with optional ControlNet+AnimateDiff photoreal rendering), with sign.mt's `pose-format` / `pose-to-video` as the open-source spine. Avoid signing-gloves detours — the literature is unambiguous they fail Deaf users.
- **Five publishable novel contributions:** (1) first NSL continuous benchmark with Deaf-community-authored annotation; (2) ISL→NSL cross-lingual transfer via YouTube-SL-25-style scraping of the NDFN 4,300-sign DVD and Pratigya Shakya dictionary videos; (3) Devanagari-aware gloss-to-text decoding with Nepali LLMs; (4) pose-based evaluation suite extending Moryossef et al.'s 2025 toolkit (arXiv:2510.07453) for South Asian SLs; (5) Deaf-co-designed avatar acceptability study comparing SMPL-X avatars vs. photoreal vid2vid for low-bandwidth Nepali deployment. Realistic 12-24 month plan: 6 Deaf NSL signers, 2 linguists, 3 ML engineers, ~\$30-60K compute, targeting ASSETS, ACL, LREC-COLING, AT4SSL, CV4A11y, CVPRW SLRTP.

1. Nepali Sign Language: Linguistic and Resource Landscape

Linguistic status. NSL (ISO 639-3 `nsp`) has been a Nepali government-recognized minority language since the 2017 Disability Rights Act. Woodward (1993) classifies NSL within the **South Asian sign-language family** alongside Indo-Pakistani Sign Language (IPSL); modern descriptive work (Hoffmann-Dilloway; Gautam 2024 in *Journal of Nepalese Studies* 16) treats it as a separate language with internal sociolinguistic variation driven by regional schools, hearing-teacher "Signed Nepali" use, and dialectal divergence from ISL. NSL phonology follows the Stokoe handshape/location/movement model (first applied to NSL by E.M. Ross). A one-handed Devanagari fingerspelling system was developed by the Kathmandu Association of the Deaf (KAD) with UNICEF support. [Medical Journal of Shree Bi...](#) [Wikipedia](#)

Population. Nepal's 2011 census reports ~80,000 people with hearing disability, but NDFN considers this an undercount and estimates ~300,000 deaf/HoH Nepalis and only ~30 certified NSL interpreters — one per ~10,000 deaf signers (NDFN Chair KP Adhikari, *Himalayan Times*).

(The Himalayan Times)

Existing NSL resources.

- **NDFN Nepali Sign Language Dictionary (2018):** ~4,300 entries, Nepali↔English glosses with sign illustrations, plus mobile app, video DVD, and deafnepal.org.np website. Developed by an NSL approval workgroup of ~20 members, with Australian Embassy support. Print volumes I-III published 1995-1997; NSL "Barhakhari" (alphabet primer) 1995. (The Himalayan Times + 2)
- **NADH dictionaries** (2003, 2014) partly absorbed into NDFN's lexicon. (Medical Journal of Shree Bi...)
- **Pratigya Shakya illustrated dictionary:** per Hoffmann-Dilloway, a "mixed descriptive and prescriptive" document. (Linguisticsociety)
- **Individual NSL educator YouTube channels** (referenced in ResearchGate 381266041).
- **Government/NGO actors:** NFDN (deaf federation, Nepal's first deaf school, 1966), NDFN, KAD, Shree Deaf School Bhairahawa, UNICEF Nepal. NDFN is a World Federation of the Deaf member. (Wikipedia)

Existing NSL computational work (essentially all of it):

- **NSL23** (Sunuwar, Borah, Kharga; Sikkim Manipal IT; *Data in Brief* 2024, doi:10.1016/j.dib.2024.110080): first published image dataset — International Manual Alphabet (A-Z), Nepali consonants/vowels/conjuncts/numbers via one-handed fingerspelling, indoor/outdoor/dark conditions. (PubMed)
- **Sentence-Level NSL Recognition** (ResearchGate 381266041, 2024): 40-sentence dataset at Shree Deaf School near Bhairahawa; MediaPipe Holistic + Spatial-Temporal Fusion Transformer; **WER 20.23%**. The only continuous NSL CSLR result. (ResearchGate)
- **NSL Character Recognition Benchmark** (Poudel, Ghimire, Bhattarai, Bhandari, Sharma Dahal; arXiv:2510.11243; *KEC Journal of Science and Engineering* 10(1), 2025/2026): 54,000 images, 36 gesture classes, 1,500/class. MobileNetV2 **90.45%** vs. ResNet50 **88.78%**. (Medical Journal of Shree Bi...)
- **Birat Poudel's** **Nepali-Sign-Language-Recognition-and-Translation** **GitHub repo:** CNN on 2,100 grayscale images/gesture, TensorFlow; community baseline code. (GitHub)

There is no continuous NSL translation dataset, no gloss-annotated NSL corpus of meaningful size, and no NSL Sign Language Production work. This is the foundational gap.

2. Foundational Dataset Construction

2.1 Lessons from major datasets

- **RWTH-PHOENIX-Weather 2014T** (Forster/Camgöz): DGS, ~9 h, ~1k vocab, 9 signers. TV weather interpreters with gloss + German subtitles. Long-standing benchmark.
- **How2Sign** (Duarte et al. CVPR 2021, arXiv:2008.08143): ASL, ~79 h, 16k vocab, 11 signers. Studio "How-To" instructional, multi-view + green-screen + depth. CMU IRB. Largest continuous ASL pre-2023, but train/test signer leakage inflates BLEU. (arxiv)
- **BOBSL** (Albanie et al.): BSL, ~1,447 h. BBC programs across many domains; subtitles weakly aligned to audio (not to signing). Honest baseline 1.00 BLEU on signer-independent topic-balanced splits. (Nips + 2)
- **OpenASL** (Shi 2022, arXiv:2205.12870): 288 h, >200 signers, scraped YouTube ASL news/vlogs.
- **YouTube-ASL** (Uthus et al., NeurIPS 2023): open-domain ASL YouTube + caption mining; 12.39 BLEU finetuned on How2Sign. (NeurIPS)
- **YouTube-SL-25** (Tanzer & Zhang 2024, arXiv:2407.11144): multilingual YouTube mining, ≥15 h/language threshold, explicitly cautions against "representation washing." (arxiv)
- **BOBSL annotation pipeline** (Momeni et al. arXiv:2208.02802): mouthing-cue + dictionary + Transformer-attention-based weak supervision at TV scale. (arxiv)
- **WLASL, MS-ASL** (isolated, YouTube), **AUTSL** (Turkish RGB-D studio), **CSL-Daily** (Chinese studio), **Content4All / SWISSTXT / VRT** (TV interpreters).
- **Indian: ISL-CSLTR** (Elakkiya & Natarajan 2021, Mendeley): 700 videos, 100 sentences, 7 signers; **INCLUDE** (AI4Bharat) isolated; **ISLTranslate** (Joshi et al. 2023, 31k pairs); **iSign** (Joshi et al. arXiv:2407.05404, ~118k video-sentence/phrase pairs, validated by ISLRTC). (Mendeley Data + 2)

Methodological lessons. (1) Three sources dominate: TV interpretation, studio with native signers, web scraping. (2) Signer-independent splits are mandatory — BOBSL is honest about 1.00 BLEU; How2Sign isn't. (3) Weak supervision works at TV scale. (4) For a continuous benchmark, target ≥20 NSL signers with regional balance (Kathmandu Valley, Pokhara, Biratnagar, Terai dialects). (5) **Deaf signers must be the authoritative annotators.**

2.2 Data format standards

- **.pose binary** (Moryossef, Müller, Fahrni 2021; github.com/sign-language-processing/pose) — Python + JS readers. Header (point count, positions, connections) + Body (frames × people × points). MIT. Adopt as canonical NSL format. (GitHub)
- **MediaPipe Holistic**: 75 upper-body + 21×2 hand + 468 face. Default in sign.mt; face landmarks noisy under occlusion.
- **OpenPose**: original 2D pose+hands+face; model hosting historically unreliable. (GitHub)
- **DWPose**: SignLLM's reprocessed Prompt2Sign (April 2025) uses it; higher hand quality.
- **RTMPose**: competitive; not yet standard in SL.

- **SMPL-X 3D**: required for avatar-quality production (Section 4).

2.3 Annotation tools

- **ELAN** (gold standard): tier hierarchy, gloss layers, non-manual annotation, frame-accurate `.eaf` integrating with the sign-language-processing toolchain.
- **iLex** (Hamburg): HamNoSys-heavy DGS workflows.
- **SignStream** (BU NCSLGR): ASL linguistic annotation.

Recommended NSL pipeline: 30 fps 1080p with two cameras (frontal + 3/4); extract `.pose` immediately with MediaPipe Holistic + DWPose dual; auto-segment with `pose-to-segments`; Deaf annotators correct boundaries and add gloss in ELAN; second-pass Deaf reviewer. Pay annotators at fair professional rates. The Volk/Müller convention is salaried Deaf teams, not crowdsourced labor.

2.4 IRB / consent

Follow How2Sign's CMU IRB protocol: (a) consent in NSL with a Deaf interpreter; (b) explicit ML training, redistribution, and withdrawal terms; (c) Bragg et al. ASSETS 2019 framework for data sovereignty.

2.5 Bootstrapping strategy for NSL

1. **License and digitize NDFN's 4,300-sign DVD** as the `WLASL-NSL` core lexicon.
2. **Scrape NSL educator YouTube channels** following YouTube-ASL methodology; apply YouTube-SL-25's 15-h-per-language minimum.
3. **Studio collection** of 1,000–3,000 Nepali sentences across health, education, government, transport, news; ≥ 20 NSL signers via NDFN.
4. **Bible/standardized translations** — JWSign (Gueuwou et al., arXiv:2311.10174) shows Bible parallel corpora as a fast on-ramp.
5. **ISL transfer** via iSign / ISL-CSLTR / ISLTranslate / INCLUDE pretraining.
6. **Synthetic augmentation**: pose-level perturbations + sign-language-processing/pose-anonymization, plus SONAR-SLT-style multilingual target augmentation.

3. Recognition Pipeline: Video → Gloss → Text

3.1 State of the art (2023–2026)

Three competitive families dominate:

- **CorrNet / CorrNet+** (Hu Lianyuy et al., CVPR 2023; arXiv:2404.11111, TPAMI 2024): pure RGB CSLR with spatial-temporal correlation + identification + temporal attention modules. **0.7% WER on CSL**, SOTA across PHOENIX-2014/2014T/CSL-Daily for both CSLR and SLT,

without pose estimation. Half the compute of CorrNet. Code at github.com/hulianyuyy/CorrNet with cross-platform demos as of Nov 2025. [\(arXiv + 4\)](#)

- **TwoStream-SLR / TwoStream-SLT** (Chen et al., NeurIPS 2022; arXiv:2211.01367): dual encoder modeling RGB + keypoints with bidirectional lateral connections, sign pyramid network, frame-level self-distillation. SOTA on PHOENIX-2014/2014T/CSL-Daily. Dominant baseline anyone cites.
- **VAC, FCN, SubUNets, CTC-based:** historical baselines; Sign Language Transformers (Camgöz et al. CVPR 2020) is the foundational joint recognition+translation transformer.
- **SignBERT / SignBERT+** (Hu, Zhao et al., TPAMI 2023; arXiv:2305.04868): self-supervised pretraining with MANO hand prior, multi-level masked modeling (joint/frame/clip). SOTA on NMFs-CSL, SLR500, MSASL, WLASL. **First self-supervised SLR framework; ideal for low-resource NSL.** [\(arXiv\)](#) [\(arxiv\)](#)
- **SHuBERT** (ACL 2025; arXiv:2411.16765): HuBERT-inspired multi-stream cluster prediction on pose-tracked hand/face/body streams. [\(arxiv\)](#)
- **Uni-Sign** (Li et al., ICLR 2025): unified SLU via language modeling, supervised pretraining on 984 h ASL + BSL, multi-task across translation and ISLR using mT5. [\(OpenReview\)](#)

3.2 Pose vs. RGB vs. multimodal

- **RGB-only:** CorrNet+ shows you don't strictly need pose with sufficient data/compute.
- **Pose-only:** cheaper, signer-invariant, privacy-preserving. Slightly lower ceiling on high-resource benchmarks but **the dominant choice for low-resource SLs** (Jiao et al. LREC-COLING 2024 achieves SOTA on PHOENIX-2014T, CSL-Daily, OpenASL, How2Sign with skeleton-only pretraining).
- **Multimodal RGB+pose:** TwoStream wins on PHOENIX. Use once NSL RGB data warrants it.

Recommendation for NSL: start pose-only with MediaPipe Holistic + DWPose dual extraction, SHuBERT-style self-supervised pretraining on all available NSL+ISL unannotated video, fine-tune with TwoStream architecture when RGB data warrants it.

3.3 Sign Language Translation

- **Cascaded Sign2Gloss2Text:** still strong (CorrNet+ on PHOENIX-2014T). [\(arXiv\)](#)
- **End-to-end gloss-free Sign2Text** (now SOTA with LLM decoders):
 - **Sign2GPT** (Wong, Camgoz, Bowden; ICLR 2024; arXiv:2405.04164): pseudo-glosses + LLM decoder.
 - **GFSLT-VLP** (Zhou et al. ICCV 2023): visual-language pretraining.
 - **GASLT** (CVPR 2023): gloss attention for gloss-free SLT.
 - **SLTUNet:** multi-task SLT with multilingual MT data.
 - **SpaMo** (NAACL 2025; arXiv:2408.10593): FlanT5-XL on PHOENIX14T/How2Sign and mT0-XL on CSL-Daily, with VT-Align training. **No visual-encoder fine-tuning required** — ideal recipe for low-resource NSL. [\(arXiv\)](#)

- **MultiStream-LLM** (arXiv:2509.00030, 2025): multi-stream (RGB+pose+mouthing+lipreading) with LLM decoder; identifies fingerspelling and non-manuals as core failure modes of single-encoder gloss-free SLT. (arXiv)
- **YouTube-ASL/T5** (Uthus et al. NeurIPS 2023) and **Zhang et al. 2024** (128 TPU-v3 $\times$ 20 days): scaling success on 6,600 h. (arxiv)
- **"LLMs are Good Sign Language Translators"** (Gong et al. CVPR 2024): quantizes sign video to LLM-compatible discrete tokens.

3.4 Devanagari / Nepali text output

- **Tokenization:** SentencePiece or **ByT5** (YouTube-ASL's best system used ByT5-XL) — byte-level handles Devanagari cleanly without Nepali-specific subword vocab.
- **LLM backbones with Nepali support:** **mT5-XL/mT0** (Muennighoff 2022); **Gemma 2/3**; **Llama 3.x with NepaliBERT/NepBERTa adapters**; **NepaliGPT** (Pudasaini et al.); **IndicTrans2 / IndicBART** (AI4Bharat). Recommendation: ByT5-large or mT5-large + LoRA on NSL-gloss $\rightarrow$ Nepali pairs.
- **Evaluation:** (sacrebleu --tokenize none) paired with (inltk) or the AI4Bharat IndicNLP Library Nepali tokenizer; BLEURT works without modification.

3.5 Self-supervised pretraining for low-resource sign

- **SignBERT+** MANO hand prior fits NSL's hand-poor data regime.
- **SHuBERT** multi-stream masked clustering — pretrain on NDFN DVD + scraped NSL YouTube without annotations.
- **BEST / SignVQNet** — discrete VQ sign tokens, transferable ISL $\leftrightarrow$ NSL.
- **Uni-Sign** mT5-based unified pretraining as the concrete recipe to follow.

4. Production Pipeline: Text $\rightarrow$ Pose $\rightarrow$ 3D Avatar (and optional Photoreal)

4.1 SLP state of the art

- **Progressive Transformers** (Saunders, Camgöz, Bowden; ECCV 2020; arXiv:2004.14874; github.com/BenSaunders27/ProgressiveTransformersSLP): first text $\rightarrow$ 3D-pose end-to-end SLP, counter decoding for variable-length output, back-translation evaluation on PHOENIX-2014T. (arXiv) (arXiv)
- **Continuous 3D Multi-Channel SLP with MDN** (Saunders 2021 IJCV; arXiv:2103.06982): mixture density networks for expressive multi-channel output and the comprehensive Deaf user evaluation that everyone cites. (arXiv Vanity)
- **G2P-DDM** (AAAI 2024): discrete diffusion for gloss-to-pose.
- **T2S-GPT** (ACL 2024): dynamic VQ autoregressive SLP.
- **SignGen** (ECCV 2024): end-to-end latent diffusion.

- **SignFlow**: Conditional Flow Matching SLP.
- **Sign-IDD** (AAAI 2025): iconicity-disentangled diffusion.
- **"Signs as Tokens"** (ICCV 2025; arXiv:2411.17799): retrieval-enhanced multilingual sign generator; beats Neural Sign Actors on How2Sign DTW.
- **"Discrete to Continuous"** (CVPR 2025): smooth transition pose generation.
- **SLRTP2025 Sign Language Production Challenge** (CVPRW 2025): target this shared task.
- **SignSparK** (arXiv:2603.10446): multilingual SLP via sparse keyframe learning; CFM in SMPL-X + MANO with WiLoR-extracted MANO and NLF SMPL-X poses; introduces a FAST sign-segmentation model. (arXiv + 3)
- **SIGNGAN** ("Everybody Sign Now"; Saunders arXiv:2011.09846): first SLP to produce photoreal sign video from spoken text. (arxiv)

4.2 The four named projects (deep technical analysis)

SignLLM (Fang et al., arXiv:2405.10718, ICCV 2025 Workshop CV4A11y; signllm.github.io). Seq2seq transformer encoder-decoder (Base/Large/Super; ~1B at Super scale) operating on 2D OpenPose/DW Pose keypoint sequences. Two modes: **MLSF (Multi-Language Switching Framework)** with parallel encoder/decoder groups per language; **Prompt2LangGloss** taking natural-language prompts. Trained with a **Reinforcement-Learning Loss + Priority Learning Channel (PLC)** for batch quality scoring. Dataset **Prompt2Sign** covers **8 sign languages: ASL, GSL/DGS, DSGS, LSF-CH, LIS-CH, LSA (Argentine), KSL (Korean), TSL (Turkish)**. **Critically: NSL and ISL are NOT in Prompt2Sign**. How2Sign back-translation BLEU-4 18.68 (dev) / 13.93 (test). License CC BY-NC 4.0. Code/data github.com/SignLLM/Prompt2Sign. Authors' stated limitations: inconsistent pose extraction across sources, costly manual gloss annotation, multilingual training cost, keypoint-only output needs external pose2vid. (arxiv + 5)

SignDiff (Fang et al., arXiv:2308.16082; signdiff.github.io). Dual-condition diffusion in pixel/latent space, **fine-tuned Stable Diffusion 1.5** on processed How2Sign frames. Introduces **FR-Net (Frame Reinforcement Network)** — a DensePose-style module enforcing text↔dense-pose-frame correspondence and suppressing multi-finger artifacts — plus a **Fast-SLP Transformer** for the text→pose stage. Trained on How2Sign and PHOENIX-2014T. How2Sign back-translation BLEU-4 17.19 (dev) / 12.85 (test); SSIM ~0.85+ vs. prior GAN methods at 0.737–0.750. **Code is being commercialized — not released**; only processed data at github.com/SignDiff/Processed-Data. Limitation: cascaded text→pose→video, single-signer training distribution. (arXiv + 3)

Stable Signer / SignX (Fang et al., arXiv:2512.04048, ICCV 2025 Demo + ACL 2026 Oral; stablesigner.github.io, signerx.github.io/SignX). Hierarchical end-to-end SLP. **SLUL (Sign Language Understanding Linker)**: T5-encoder-based Prompt2Gloss/Text2Gloss with **SAGM (Semantic-Aware Gloss Masking) Loss** + KL-divergence + contrastive losses. **SLP-MoE**: Mixture-of-Experts hand-gesture rendering with rule-based stable poses + diffusion-style rendering. How2Sign back-translation: Full Stable Signer **BLEU-4 23.24 (dev) / 15.67 (test)**, beating SignLLM by ~48.6% BLEU-4 (dev) / 63.7% (test) and beating Neural Sign Actors' test BLEU-4 (13.12) by a wide margin. SLUL Prompt2Gloss alone: test BLEU-4 30.74, ROUGE 69.72

(+43.3%/+23.5% over SignLLM Text2Gloss). **Code will not be made public ("commercialized")**. Demos in the same 8 sign languages as SignLLM; **NSL/ISL absent**. (arxiv + 7)

Neural Sign Actors (Baltatzis et al., CVPR 2024; arXiv:2312.02702; baltatzisv.github.io/neural-sign-actors). Text-conditioned **diffusion in SMPL-X parameter space** (not image, not vertex). Denoiser: anatomically-informed pose+expression encoders built on a **GNN defined on the SMPL-X body skeleton** (breaks MLP permutation-equivariance) + CLIP text encoder + autoregressive LSTM decoder. Hand-pose loss weighted $\times 2$. Trained on a new 3D ASL dataset built by fitting SMPL-X to How2Sign with OSX + MediaPipe hand refinement + temporal coherence loss; beats SGNify on fitting. SOTA at publication time on MPJPE/FID/DTW/BLEU/ROUGE. Project page promises code release but **no public GitHub repo currently**. Limitations: SMPL-X expression channel only partially captures facial non-manuals; depends on automated 3D fitting quality; ASL-only. (arXiv + 6)

SignAvatars (Yu et al., ECCV 2024; arXiv:2310.20436; github.com/ZhengdiYu/SignAvatars). First large-scale 3D SL holistic-motion dataset: **8.34M SMPL-X annotations across 70K motion sequences from 153 signers**, with MANO. Four sub-datasets: ASL (How2Sign), GSL (PHOENIX14T), HamNoSys (isolated), Word (WLASL). License **non-commercial, gated by Google Form**; RGB videos not redistributed (license reasons). **Most valuable 3D pretraining substrate available**. (ECVA + 2)

4.3 3D body models

- **SMPL-X** (Pavlakos et al., CVPR 2019): SMPL + MANO hands + FLAME face. Used by Neural Sign Actors, SignAvatars, DexAvatar, SignSpark. (arXiv)
- **MANO** (Romero et al.): hand-only; SignBERT+ uses it as a hand prior.
- **FLAME** (Li et al.): face/expression model; essential for grammatical non-manuals (eyebrow raise → question; lip rounding → adverbial; mouthing).
- **Why these matter:** In NSL as in every SL, facial expression and mouthing carry **grammatical** weight (Bridges & Metzger 1996; Brentari 2010). Skeleton-only keypoints cannot represent mouth shape, eyebrow position, cheek puff. Body-keypoint-only production (e.g., SignLLM) is *grammatically deficient*; **SMPL-X is the minimum acceptable target for production-quality avatar SLP**.

4.4 Rigged avatar approaches

Approach	Strengths	Weaknesses for sign	License
Mixamo rigs (in pose-to-video repo)	Free, plug-and-play, Blender-compatible	Limited hand articulation; no facial blendshapes	Free (Adobe)
Blender + Rigify / Auto-Rig Pro	Free, full control, FACS-capable face rigs	Manual rigging, slow iteration	GPL

Approach	Strengths	Weaknesses for sign	License
Unreal MetaHumans	Photoreal, MetaHuman Animator (ARKit), free	Unreal-only; uncanny valley	Unreal EULA, free for many uses
Ready Player Me	Easy customization, web SDK	Limited finger control; not sign-designed	Commercial SaaS
VRoid	Anime-style, fast	Anime aesthetics may not suit; limited fingers	Free
JASigning / SiGML (Ebling & Glauert)	SL-specific, used in Swiss DGS Deaf community studies	Older toolchain, dated visuals	Academic

For NSL: train a SignAvatars-style SMPL-X pose generator; render via **Blender + SMPL-X add-on** with Deaf-validated NSL non-manual blendshapes; provide *optional* photoreal via the pose-to-video stack (ControlNet + AnimateDiff over Stable Diffusion 1.5/SDXL). Stylized initially to avoid uncanny valley.

4.5 Signapse.ai (what they actually do)

UK-based; Crunchbase shows obfuscated total funding across 6 rounds. Two products:

Crunchbase

- **SignStudio**: post-production translation (video/audio/PDF → BSL/ASL), Webex App Hub integration, both AI and human translator paths. Tiered (Starter/Team/Enterprise), 12-month minimum commitment, monthly billing. Webex App Hub + 2
- **SignStream API**: real-time BSL/ASL from English text, low-latency for live events, broadcast overlays, digital signage. Explicitly disclaims: "entirely AI-generated, with no human grammar checking or quality assurance ... not suitable for formal, legal, or official use." Crunchbase Signapse
- Architecturally inferable: a finite-vocabulary text→gloss→video-clip-concatenation system using a library of pre-recorded signer ("Rae") clips, blended for transitions. Public **Sign Language Production Framework (SLPF)** roadmap targets Tier 7; QA via Deaf BSL reviewers scoring clarity, grammar, and expression. SignStudio guarantees 100% understandability via human review; SignStream does not. Signapse Signapse
- A sensible commercial architecture, but fundamentally **lookup-and-stitch**, not generative. Your NSL system, if generative (Stable Signer / Neural Sign Actors line), would be technically *more* novel than Signapse.

4.6 vid2vid for photoreal rendering

- **vid2vid** (Wang et al., NVIDIA, NeurIPS 2018): original pose→video GAN, signer-specific.
- **vid2vid-cameo**: few-shot adaptation for video calls.
- **EDN** ("Everybody Dance Now," Chan et al.): pose→video body baseline.

- **Pix2PixHD**: in pose-to-video repo with sign.mt-hosted weights. (GitHub)
- **SIGNGAN** (Saunders arXiv:2011.09846): first photoreal SLP from spoken text.
- **ControlNet + AnimateDiff**: pose-controlled diffusion, the current preferred photoreal stack in (sign-language-processing/pose-to-video). (GitHub)
- **SignDiff**: SD 1.5 fine-tune with FR-Net.
- **Signer Avatar** (arXiv:2508.15988, 2025): 3D dilated convolutions + Sapiens features for demographically diverse photoreal signers. (arxiv)

4.7 Diffusion / motion-generation toolkits

MDM (Tevet et al., ICLR 2023), **MotionDiffuse** (Zhang et al. 2022), **T2M-GPT** (Zhang et al. CVPR 2023), **MotionGPT**, supporting datasets **HumanML3D** / **KIT-ML**. 2024–2026 trend: import wholesale and add hand/face priors. Neural Sign Actors does exactly this with the SMPL-X-GNN denoiser.

4.8 Avatar vs. photoreal: what Deaf users prefer

- **Adamo-Villani et al. 2015**: stylized and photoreal had *equivalent legibility*, but stylized rated significantly more *appealing* (uncanny valley effect). (Frontiers)
- **Brock et al. 2018**: ASL users preferred natural-style over anime-style. (Frontiers)
- **Quandt et al., Frontiers in Psychology 2022** (doi 10.3389/fpsyg.2022.730917): attitudes vary by hearing status, age of SL acquisition, and avatar type; Deaf-of-Deaf users have higher tolerance for non-photoreal.
- **SIGNGAN** (Saunders 2020) argues prior graphical avatars "have proven unpopular" while skeleton outputs are "not understandable to Deaf viewers" — motivating photoreal. Contested. (arxiv)
- **Signapse's photoreal "Rae"** suggests B2B will pay for photoreal; **Hand Talk's stylized "Hugo"/"Maya"** (selected by the UN as the world's best social app at WSA-mobile 2013 in Abu Dhabi; over 9 million downloads and 1 billion+ translated words across Libras/ASL/BSL per handtalk.me) shows stylized works at scale for free consumer education. (Whoisaccessible) (Google Play)

For NSL: support both — stylized SMPL-X avatar for the open-source/education product; optional photoreal pose-to-video for commercial deployments paying for realism premium.

5. Evaluation Metrics and Methodology

SLR: WER (PHOENIX/CSL standard; CorrNet+ reaches 0.7% on CSL); gloss-level accuracy for isolated SLR. (arXiv)

SLT: BLEU-1..4 (standard but flawed — BOBSL 1.00 BLEU on signer-independent splits despite 1,447 h); ROUGE-L; METEOR; **chrF/chrF++** (good for Devanagari/morphologically rich Nepali);

BLEURT (correlates better with humans); COMET/COMET-QE *cannot* be used source-based for sign→text (no spoken-language source); **SignBLEU** (Kim et al. 2024) — multi-channel sign-output BLEU splitting manual/non-manual signals into separate temporal channels. (NeurIPS)

SLP:

- **Back-translation BLEU** — train a separate SLT on the same data, run generated poses through it, score against reference text. Standard since Saunders ECCV 2020. Pitfall: sensitivity to SLT backbone — fix by using a frozen, well-documented one. (arXiv)
- **DTW on keypoints/pose** — standard distance; Moryossef et al. 2025 **pose-evaluation toolkit** (arXiv:2510.07453) provides nMSE, nAPE, and nDTW-MJE with shoulder-joint normalization and low-confidence-joint masking. (arXiv)
- **FID on rendered video** for photoreal.
- **MPJPE** for SMPL-X SLP.

Why human evaluation is essential. Automatic metrics are systematically misleading. Moryossef et al. 2025 finds keypoint-distance metrics (well-tuned) and back-translation likelihood correlate best with Deaf-rater ratings; embedding-based metrics fail at sentence level. WMT-SLT22 and WMT-SLT23 both required Deaf-signer human evaluation as the official ranking criterion. Tarrés et al. (CVPRW 2023) warn on How2Sign: "the common practice of choosing the best model checkpoint based on the BLEU metric may be misleading. ... the model tends to capture frequent patterns, and may even provide realistic outputs completely unrelated to the input video." (arXiv) (arxiv)

Practical Deaf-rater design for NSL (per WMT-SLT and Saunders): ≥10 Deaf NSL signers balanced by age/sex/region; blind A/B vs. reference; rate clarity, grammar, naturalness, comprehensibility (4-point Likert each) + free-text NSL video comment; include "creepiness" rating for avatar work (Quandt 2022); pay at fair NSL-professional rates (\$25-50/hour minimum), formalized via NDFN.

6. Novel Research Contributions

Five contributions a Nepali-led project can credibly target at top venues:

1. **"NSL-Bench": the first continuous Nepali Sign Language benchmark with Deaf-co-authored annotation and signer-independent splits.** BOBSL-style splits, not How2Sign-style leaky splits. Co-designed with NDFN, the Pratigya Shakya estate, and Shree Deaf School Bhairahawa. **Target: LREC-COLING 2026/2027 (resource), ACL 2026 (resource), NeurIPS Datasets & Benchmarks 2026.** Highly defensible — no NSL continuous dataset exists.
2. **Cross-lingual ISL→NSL transfer using pose-level pretraining.** Replicate YouTube-SL-25's multilingual transfer finding for NSL specifically: pretrain on pooled ISL (iSign 118k, ISLTranslate 31k, INCLUDE, ISL-CSLTR) + scraped NSL video using SHuBERT-style multi-

stream masked pretraining; fine-tune the supervised gloss-free SLT head. **Target: ACL Findings 2026, EMNLP 2026, CV4A11y ICCV 2027.** (arXiv)

3. **Devanagari-aware gloss-to-text decoding for Indo-Aryan SLs.** Most SLT decoder work assumes Latin-script targets; even YouTube-ASL's ByT5 setup hasn't been systematically studied on Devanagari. Compare ByT5, mT5, IndicTrans2, NepaliGPT, and Llama 3.x + LoRA on identical NSL gloss→Nepali pairs. Report Nepali-specific tokenization effects on BLEU/chrF. **Target: EMNLP 2026, NAACL 2026, AT4SSL 2026.**
4. **"Pose-NSL-Eval": pose-based evaluation toolkit extension for Indo-Aryan sign families.** Build on Moryossef et al. 2025's pose-evaluation library, adding NSL-specific reference test sets and SignBLEU-style multi-channel metrics adapted for NSL's non-manual inventory. **Target: AT4SSL 2026, LREC-COLING 2026, ASSETS 2026.**
5. **"Avatar Acceptability in the Global South": a Deaf-co-designed comparative study of rigged SMPL-X avatars vs. photoreal vid2vid output, conducted with NSL signers across rural and urban Nepal.** All existing avatar-acceptability literature (Adamo-Villani, Brock, Quandt, Kipp, Krausneker) is from the Global North. The Global South dimension — bandwidth constraints, mobile-first deployment, culturally specific stylization preferences — is uncharted. **Target: ASSETS 2026, CHI 2027, *Universal Access in the Information Society*.**

Honorable mention angles: a first explicit "NSL grammar breaks model X" stress test (one-handed Devanagari fingerspelling, NSL classifier predicates, Nepali honorific encoding via non-manuals); production-quality NSL→Nepali → low-resource Nepali-Bhojpuri/Maithili cascaded SL→spoken-LRL pipelines.

Venues to monitor and submit to: CV4A11y at ICCV/CVPR, AT4SSL (EAMT), SLTAT (Sign Language Translation and Avatar Technologies), WMT-SLT shared tasks, **ACL accessibility track**, ASSETS (ACM SIGACCESS), SLRTP Sign Language Production Challenge at CVPRW.

7. Open-Source Tooling and Reference Implementations

7.1 The sign-language-processing GitHub organization (github.com/sign-language-processing)

Run by Amit Moryossef and collaborators; the most coherent open-source SL stack and what powers **sign.mt** (the live demo) and **research.sign.mt** (the literature overview):

- **pose-format** — Python + JavaScript readers/writers for the **.pose** binary format. **MIT.** Adopt as canonical NSL data format. (GitHub)
- **pose** — viewing, augmenting, normalizing. **MIT.**
- **pose-to-video** — pix2pix (sign.mt-hosted weights), ControlNet (sign/sd-controlnet-mediapipe), AnimateDiff, StyleGAN3, Mixamo. **MIT.** (GitHub)
- **pose-anonymization** — de-identify pose sequences. **MIT.** (GitHub)

- **spoken-to-signed-translation** — text→gloss→pose→video, glossers (**simple**), **spacylemma**, **rules**, **nmt**), lexicons (SignSuisse for DSGS/LSF-CH/LIS-CH; WordNet "Coming Soon" for GSG and BSL). Published at AT4SSL 2023. [GitHub](#) [GitHub](#)
- **segmentation** — sentence- and sign-level BIO tagging with optical-flow inputs. [GitHub](#)
- **transcription** — end-to-end pose→SignWriting/HamNoSys→text. [GitHub](#)
- SignWriting toolchain: **signwriting-translation**, **signwriting-transcription**, **signwriting-description**, **signwriting-evaluation**, **signwriting-animation**. [GitHub](#)
- **fluent-pose-synthesis** — post-edit existing SL videos by recording corrections and diffusing transitions. [GitHub](#)
- **recognition** — lexical sign recognition. [GitHub](#)
- **datasets** — TFDS loaders. [GitHub](#)

All MIT-licensed, designed to be assembled. Use this as the NSL spine; add NSL lexicons, NSL glossers, NSL pose checkpoints.

7.2 sign.mt

Web interface at sign.mt; architecture overview at research.sign.mt. References Pakistan Sign Language via SpreadTheSign lexicon. **NSL is not currently supported** — adding it is a concrete contribution to the open ecosystem. [GitHub](#) [GitHub](#)

7.3 SignLLM / Prompt2Sign

github.com/SignLLM/Prompt2Sign. CC BY-NC 4.0. DWPose-extracted preprocessed data; tools under **tools-new-2025/**. Useful as a benchmark target and multilingual pretraining substrate, but its non-commercial license precludes commercial NSL product use. Plan around this.

7.4 Google sign-language research

- **PopSignAI** — ASL vocabulary game with classifier; PopSign ASL v1.0 dataset.
- **SignTown** — older AR demo.
- **MediaPipe Holistic** — the de facto pose extractor.
- **YouTube-ASL / YouTube-SL-25** — datasets and baselines (NeurIPS 2023, 2024).
- **FLEURS-ASL** — multilingual benchmark. Code typically Apache-2.0; data under YouTube ToS (research-use).

7.5 Meta sign research

Less public than Google's. HumanML3D-adjacent motion generation is relevant but not specifically signed.

7.6 Hugging Face

Search "sign language" yields ControlNet checkpoints (**sign/sd-controlnet-mediapipe**), SignBERT checkpoints, How2Sign loaders, several university models. Mixed Apache-2.0 / CC-BY-NC — read each model card.

7.7 SignAvatars

github.com/ZhengdiYu/SignAvatars. Code MIT; data **non-commercial, gated by Google Form**. Single most useful 3D pretraining substrate.

7.8 Others

OpenHands (Kezar, Thomason, Sehyr): phonology-aware ISLR integrating ASL-LEX 2.0. **Sign-Language-Datasets** (Moryossef): TFDS adapters. **ProgressiveTransformersSLP** (Saunders): canonical SLP baseline. **CorrNet / CorrNet+** (Hu Lianyuy).

8. Commercial Landscape and Productization

8.1 Competitive landscape

Company	Languages	Approach	Notable verticals
Signapse.ai (UK)	BSL, ASL	Photoreal lookup-and-stitch "Rae"; SignStream API + SignStudio	UK rail (GoMedia "Luna" product powered by Signapse, RailAdvent) trialed Jul-Sep 2024 at five London Overground stations — Hackney Central, Hackney Downs, Hoxton, Upper Holloway, Willesden Junction — operated by Arriva Rail London for TfL, funded by UK DfT First of a Kind 2024; Icomera broadcast media; signage
Hand Talk (Brazil)	Libras, ASL, BSL	Stylized 3D "Hugo"/"Maya"; statistical MT Gcppodcast + Firebase backend	EY, Avon (web plug-in); ImpactAlpha 9M+ app downloads; selected by the UN as world's best social app, WSA-mobile 2013 Abu Dhabi; Hand Talk 1 of 20 organizations sharing the \$25M Google AI Impact Challenge in May 2019
SignAll (Hungary/USA)	ASL	4-camera (incl. depth) + NLP; Nanalyze Gallaudet partnership Wikipedia Gallaudet University	Education, government
KinTrans (Dallas)	ASL, Arabic, more planned	3D camera (Kinect) → text/voice GOOD + back-avatar; Microsoft Azure Microsoft	UAE banks, government offices RWS

Company	Languages	Approach	Notable verticals
SLAIT.ai	ASL	Real-time SLR for video calls	B2B
Silence Speaks / Madduru (UK)	BSL, ASL	Photoreal avatar with QR-code workflow (MultiLingual) (separate company; not the GoMedia/Signapse rail product)	Real-time announcements at >250 train stations announced; greeting cards (MultiLingual)

Failed/criticized approaches: SignAloud gloves (University of Washington, 2016) and similar — uniformly criticized by the Deaf community (Audio Accessibility, Language First, ASHA Leader) for: (a) one-way translation only, (b) ignoring non-manual grammatical signals, (c) not consulting Deaf signers, (d) reinforcing the burden-on-Deaf-person framing. **Do not pursue glove-based or wearable-sensor NSL.** (RWS + 6)

8.2 What a commercial NSL product would look like

Highest-value Nepal verticals:

- Government services** — Ministry of Health, Department of Transport: live announcement signing in train/bus stations; government office kiosks. Mimics the TfL/Signapse model.
- Education** — NSL learning app modeled on Hand Talk; freemium + B2B school licenses. Nepal's deaf-education infrastructure is small but concentrated (22 government-run schools for Deaf students and 172 resource classrooms, totaling roughly 194 settings, per Kathmandu Post (Jun 23, 2019) quoting NDFN President KP Adhikari and corroborated by Humanity & Inclusion at hi-us.org) — a captive market for a single quality product.
(Kathmandupost) (Kathmandupost)
- Healthcare** — clinic kiosks for triage / patient information, with hard guardrails ("not for diagnosis without human interpreter").
- News broadcasting** — partner with Nepal Television, Kantipur TV for evening news signing.
- Transport** — Tribhuvan Airport, Kathmandu Metropolitan buses.
- Banking** — Nabil, NIC Asia teller kiosks (paralleling KinTrans UAE).

Pricing models: B2B SaaS subscription (Signapse-style, NPR-denominated monthly tiers); B2C freemium app (Hand Talk-style); B2G fixed-fee contracts via NDFN partnership.

8.3 Funding sources for Nepal / South Asia accessibility tech

- World Bank Disability Inclusion Initiative** — direct grants in LMICs.
- USAID** — Nepal mission has historically funded accessibility (2025+ policy status: verify before assuming).

- **Google.org Accessibility** — funded Hand Talk in the May 2019 Google AI Impact Challenge cohort and has funded PopSignAI.
 - **Microsoft AI for Accessibility** — funded WMT-SLT human evaluation, KinTrans early stages.
 - **GAATES** — Global Alliance on Accessible Technologies and Environments.
 - **Asian Development Bank (ADB)** — disability inclusion programs in South Asia.
 - **NORAD** (Norway) and **DFAT** (Australia, the existing NDFN dictionary funder) — bilateral aid windows.
 - **Nepal CSR** — Ncell, Nabil, NIC Asia, NTC ESG mandates.
 - **Academic:** SNSF (Switzerland; precedent in SMILE/Content4All), Horizon Europe accessibility calls (with an institutional EU partner).
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9. Ethical and Community Considerations

9.1 "Nothing about us without us"

This is not a slogan; it is the empirical constraint that separates successful from failed SL AI projects. Every long-running successful effort has Deaf co-founders or co-PIs:

- **Hand Talk:** Deaf engineers from inception (Tenório / Madduru collaborators).
- **Signapse:** "team of Deaf and hearing people; entrepreneurs, engineers, and researchers." [Crunchbase](#)
- **SignAll:** developer is Deaf; team includes Deaf and hearing engineers and linguists from Gallaudet. [Signall](#) [Wikipedia](#)
- **iSign:** validated by ISLRTC; content from Deaf-led YouTube channels (ISH-News, DEAF ENABLED FOUNDATION). [Exploration-lab](#)
- **Failed projects** (signing gloves): hearing-only inventors, no Deaf co-design. [The ASHA Leader](#)

Operational implication: Deaf NSL signers as **co-investigators**, not "informants" or "consultants." This affects authorship on papers (Deaf signers as co-authors), equity in any spinout, and decision authority over data release.

9.2 Past failed projects

- **SignAloud gloves** (UW, 2016): Lemelson-MIT prize → community backlash. Lessons: (a) ASL is not fingerspelled English; (b) gloves miss 50%+ of grammar carried by face/body; (c) one-way only; (d) "savior" framing. [Language First](#) [Language First](#)
- **EnableTalk, Acceleglove, SignAloud Gloves 2.0:** all variants, all failed.
- **Mexico "Glove Translator" 2015:** wrong signs in promotional video; offensive "deaf and mute" framing. [Audio Accessibility](#)
- **MotionSavvy:** better-intentioned (Deaf RIT/NTID students) but limited by Leap Motion hand-only tracking. [Wikipedia](#)

- **Avoid:** hardware-first thinking; treating SL as gestures; building only for one direction.

9.3 Data sovereignty for Deaf communities in the Global South

- **NDFN holds redistribution rights**, not the researchers' university.
- **CC BY-NC-SA** for the open-source NSL benchmark; commercial-use licenses negotiated through NDFN.
- **Signer images may be withdrawn** at any time (right of erasure).
- **Pose-only release option** is dignity-preserving (**pose-anonymization** tooling).
- Follow **Bragg et al. 2019 ASSETS** framework and the **Te Mana Raraunga** indigenous-data parallel.

9.4 Critical-setting deployment risk

Signapse explicitly warns SignStream is "not suitable for formal, legal, or official use." Adopt the same posture for NSL: **Signapse**

- **Never deploy automated NSL translation in medical (diagnosis, consent), legal (court, contracts), or emergency (hospital triage, police) settings without a human NSL interpreter as primary.** Automation supplements, never replaces, in critical settings.
 - Build clear UX warnings into the product (Hand Talk and Signapse both do).
-

10. Concrete Phased 12-24 Month Roadmap

10.1 Team composition

- **2 Deaf NSL signers as co-PIs:** one NDFN-affiliated, one with linguistics training.
- **4-6 Deaf NSL annotators** (paid, salaried).
- **1 NSL linguist** (e.g., Tribhuvan University; potentially Bhuvan Gautam or affiliated descriptive linguist).
- **3 ML engineers:** 1 vision/pose lead, 1 NLP/LLM lead, 1 SLP/avatar lead.
- **1 PM / community liaison** (Deaf or NSL-fluent).
- **1 Blender/3D artist** (rigging, SMPL-X integration).
- **Advisory board:** NDFN leadership; an ISL researcher (e.g., iSign team); a sign-language-processing org collaborator (e.g., Moryossef); an accessibility legal advisor.

10.2 Compute

- **SSL pretraining** (SHuBERT-style on ~500-2000 h pooled ISL+NSL): 8× A100 80GB for 2-4 weeks ≈ **\$15-30K** on cloud (or 1× DGX-class on-prem).
- **SLT fine-tuning** (ByT5/mT5 + visual encoder): 4× A100 for 1-2 weeks ≈ **\$3-6K**.
- **SLP 3D diffusion** (NSA/SignSpark style on NSL): 8× A100 for 2 weeks ≈ **\$10-15K**.

- **Inference / demo serving:** 1× A10G or L4 cloud GPU, ~\$300/month.
- **Total Year-1 compute:** ~\$30–60K (Lambda Labs, Crusoe, or similar).

10.3 Phased milestones

Phase 0 — Foundations (Months 0–3): formal MoU with NDFN (data rights, IRB, salaried Deaf staff); recruit Deaf co-PIs and annotators; set up infrastructure (`ns1-language-processing` GitHub org, `.pose` pipeline, ELAN environment, NSL-video-friendly comms); digitize NDFN 4,300-sign DVD and Pratigya Shakya dictionary with NDFN consent and clear redistribution license; IRB at host institution + NHRC (Nepal Health Research Council) clearance. **Deliverable:** signed data agreement, ~5k NSL sign videos ingested, internal data dashboard.

Phase 1 — Dataset Construction (Months 3–9): Pose-extract NDFN DVD + scraped NSL YouTube + studio sessions (target ≥ 20 NSL signers, regional balance, ≥ 30 hours continuous + 5,000 isolated signs). Deaf-led ELAN with double-pass review. BOBSL-style signer-independent topic-balanced splits. Release **NSL-Bench v0.1** under CC BY-NC-SA via NDFN. **Paper #1 target:** LREC-COLING 2027 resource track or ACL 2027 resource track, co-authored with Deaf NSL signers and NDFN representatives.

Phase 2 — Recognition Pipeline (Months 6–15, overlapping): SHuBERT-style multi-stream SSL pretraining on pooled ISL (iSign 118k, ISLTranslate 31k, INCLUDE, ISL-CSLTR) + NSL scraped video. Supervised fine-tune: pose-only baseline (CorrNet+ ported to pose features) + multimodal TwoStream-SLR. Gloss-free SLT with ByT5-XL / mT5-large SpaMo-style VT-Align vs. cascaded Sign2Gloss2Text with IndicTrans2 decoder. Devanagari-aware tokenization study; report BLEU, chrF, BLEURT, and Deaf-rater human evaluation. **Paper #2** at ACL/EMNLP/NAACL Findings 2027 (Devanagari decoding); **Paper #3** at CV4A11y ICCV 2027 (ISL→NSL transfer).

Phase 3 — Production Pipeline (Months 9–18): Fit SMPL-X to NSL data following SignAvatars / Neural Sign Actors methodology (OSX + MediaPipe-refined hands + temporal coherence loss). Train an NSA-style diffusion + SignSparK-style keyframe CFM on NSL. Render via Blender + SMPL-X add-on with Deaf-validated NSL non-manual blendshapes. Optional: photoreal pose-to-video via ControlNet + AnimateDiff on a single NDFN-approved signer. Deaf-co-designed avatar acceptability study (rural + urban Nepal, ≥ 30 evaluators). **Paper #4** at ASSETS 2027 (avatar acceptability); **Paper #5** at CVPR/ICCV 2027 (NSL SLP method with ISL transfer + SMPL-X target).

Phase 4 — Open-Source Prototype Release (Months 15–18): Release (`ns1-language-processing` GitHub org with pose checkpoints, gloss-free SLT model, NSL avatar pipeline, Blender add-on, and web demo (modeled on sign.mt). Upstream NSL lexicon/translator into sign.mt. HF checkpoints under CC BY-NC-SA (commercial via NDFN).

Phase 5 — Commercialization Track (Months 18–24): Spin out a Nepal-incorporated PBC under Nepal's 2017 Social Enterprise framework with NDFN as co-founder/equity-holder. Pilot deployments at Tribhuvan Airport announcements; Nabil Bank teller kiosks; Nepal Television evening news signing; Ministry of Health emergency-alert signs. Apply for Google.org /

Microsoft AI for Accessibility / World Bank / ADB grants. SaaS dashboard (Signapse-style) priced in NPR for local SMBs. Reserve commercial-license revenues per agreement with NDFN.

10.4 Target paper venues by year

- **Year 1:** ASSETS 2026 (community/protocol), LREC-COLING 2026 (resource), AT4SSL 2026 (eval methodology), WMT-SLT 2026 shared-task submission.
- **Year 2:** ACL/EMNLP 2026/2027 Findings (decoding), CV4A11y ICCV 2027 (transfer), CVPRW SLRTP 2026 (SLP challenge submission), CVPR/ICCV main 2027 (full system or avatar paper).

10.5 Risk register and pivots

- **NSL data scaling slower than planned** (<20 h in Phase 1): pivot to a high-quality 5–10 h benchmark + heavier ISL transfer; reframe Paper #1 as a "low-resource SL methodology" paper rather than a flagship dataset paper.
- **Avatar acceptability strongly favors photoreal:** shift the open-source product to a hybrid (stylized default + photoreal opt-in) and partner with a single NDFN-approved photoreal signer.
- **Competing commercial NSL effort emerges** (e.g., a Hand Talk Nepal launch): differentiate via open-source spine, NSL-native team, government-services vertical.
- **Devanagari LLM decoding underperforms:** add a rules-based Nepali morphological post-processor (IndicNLP Library) and keep the cascade Sign2Gloss2Text path alive as backup.

10.6 Hard "no-go" thresholds (kill criteria)

- If after 12 months you cannot deliver a Deaf-co-authored peer-reviewed paper and an open-source benchmark under NDFN partnership: **pause** the productization track. Open-source NSL research without Deaf co-authorship is the failure mode you must avoid.
- If a Deaf-rater human evaluation gives the production avatar <2/4 on naturalness in two consecutive iterations: do not deploy to government/transport pilots; restrict to consumer education uses only.

Caveats

1. **NSL linguistic descriptions in this report rely on Woodward 1993, Hoffmann-Dilloway, and Gautam 2024;** no independent corpus-linguistic validation of NSL grammar is currently available at scale. A modern descriptive grammar of NSL is itself a research need.
2. **Signapse, Hand Talk, KinTrans, and SignAll technical descriptions** are inferred from public marketing, press releases, podcast interviews, and adjacent academic work; none have published full architecture papers. Treat as approximations.

3. **SignDiff, SignLLM, Stable Signer/SignX BLEU numbers** are self-reported in arXiv preprints, several with closed-source code. Independent reproductions are limited. The "+48.6% BLEU-4" Stable Signer figure is a *relative* improvement vs. the Fast-SLP-Transformer baseline (SignDiff family), not an absolute SOTA across all benchmarks.
4. **NDFN's reported ~300,000 deaf population and ~30 interpreters** is NDFN's own estimate versus the 2011 census ~80,000; the truth lies between, and a 2021/2026 census update should be checked.
5. **Funding sources change:** USAID Nepal mission status in 2025–2026 is in flux; verify availability before assuming.
6. **The recommendation to use IndicTrans2 / NepaliGPT / Llama 3.x assumes 2025–2026 model availability;** new Devanagari-capable LLMs (Gemma 3, Qwen 3, etc.) are emerging quickly. Re-evaluate at Phase 2 start.
7. **All license claims** (CC BY-NC for SignLLM/SignAvatars; MIT for sign-language-processing; closed for SignDiff/SignX code) reflect public statements as of late 2025; re-verify at integration time.
8. **The roadmap deliberately omits "build an LLM tutor for NSL learners" and other plausible adjacent products** — these are valuable but dilute focus; revisit in Year 3.